

Andreas O'Brien

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Professional Profile

Data analyst with cross-domain experience in renewable energy analytics, cloud data workflows, and customer-facing delivery. Skilled in translating operational and business requirements into scalable Microsoft Fabric and Power BI solutions, with hands-on capability in Python, SQL, DAX, and PySpark. Combines a strong scientific foundation (B.Sc. in Nanoscience, Physics and Chemistry of Advanced Materials) with applied data science training (Postgraduate Diploma in Data Science). Recognised for clear stakeholder communication and effective presentation of analytical insights to technical and non-technical audiences alike. Native English and German speaker.

Education

Data Science Institute, Ireland (2024)

Postgraduate Diploma in Data Science

- Formalized industry data science experience across EDA, visualization, and machine learning.
- Final project: machine-learning-based stock trading algorithm.

Trinity College Dublin, Ireland (2015-2019)

B.Sc. in Nanoscience, Physics and Chemistry of Advanced Materials

- Interdisciplinary training in physics, chemistry, and mathematics, with laboratory and programming focus.

Ardcoil Ris Secondary School, Limerick, Ireland (2009-2015)

Leaving Certificate: 615 points

- Physics (A1), Chemistry (A1), Maths (A1), Applied Maths (A1), German (A1), Music (A2)

Certifications (CPD)

- Microsoft Certified: Power BI Data Analyst Associate (PL-300)
 - Generating visuals in Power BI to support rapid, intuitive analysis for end users.
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)
 - Planning, implementing, and maintaining analytics solutions.
 - Experience with data ingestion, transformation, storage, and analysis using SQL, DAX, and PySpark.

Work Experience

Data Analyst, Applied Technologies (October 2025-Present)

- Enabling **renewable energy** companies in adopting **Microsoft Fabric** based cloud workflows.
- Building and maintaining **data pipelines**, **data models**, and **reporting layers** to improve analytics delivery and decision-making.
- Translating business and operational requirements into scalable Fabric solutions across **ingestion**, **transformation**, and **visualization**.

Energy Analyst, EM3 Energy (November 2023-October 2025)

- Worked in a multidisciplinary role bridging **energy engineering** and **data science** functions.
- Developed advanced **energy models** for baseline energy consumption and performance comparison.
- Delivered analysis across energy consumption, financial forecasting, and energy audits.
- Developed proficiency in **SQL** and **Power BI**.
- Held a **customer-facing** role delivering projects from **concept to deployment** for large energy consumers in the EU and Asia.

Assistant Researcher, RWTH Aachen (June 2020-March 2022)

- Worked at the **Surface Engineering Institute** (IOT - RWTH Aachen).
- Investigated particle temperature profiles in thermal spray processes using ultra-high-speed imaging and novel computational analysis methods.
- Used **Python** and **MATLAB** to process **large datasets**.
- Published findings in the Journal of Thermal Spray Technology.

Technical Skills

- Programming and analytics: Python, SQL, DAX, PySpark
- Data visualization and BI: Power BI, Microsoft Fabric
- Domain knowledge: Energy modelling, time-series forecasting, wind farm data processing

Languages

- English (native)
- German (native)

References

References available on request.